v4 — ignoring the clutter clears the port strip

V4_CHECK.md scored rule B on every known dry pixel and stopped. Kop van Zuid passed. The port strip did not: layover minus nominal was +1.18 / +1.04 dB in the near and middle thirds, and the bar was +1.5. The same labels, scored only away from clutter, had been +1.94 / +3.64. The shortfall was the port. Container stacks, cranes, ships and trees are not in a buildings-only label, so their bright returns sat in the nominal class and pulled the contrast down.

This note does not move the roof. Rule B (`image_space.own_range_grid`), the buildings-only scene, and the AHN4 DTM ground stand. There is no roof-nominal edit. What changes is where a label is claimed. Mask value 255 means the pixel is ignored: it is out of the loss, the class weights, every score, and the brightness baseline. v2 was not modified. The v2 image, incidence, look, grid, split and orientation are copied. `label_version` is `v4-inplace-dtm-ign`.

Ignore

On the MAG-POL grid, in canonical orientation (sensor at row 0):

bit	what it marks
(a)	the projection said unknown (255), as before
(b)	nDSM > 3 m, DTM ground, outside 3DBAG footprints dilated 1 m, then dilated another 2 m (8 px)
(c)	the buildings-only label and the full-DSM label disagree, both from `own_range_grid` with the DTM as the ground

Water is not in (b) or (c). (a) is empty on every strip that was measured: the projection no longer leaves unknown pixels once the DTM fills the ground.

strip	ignored	(a)	(b) clutter	(c) full ≠ bldg	(b) only	(c) only	water
P `...22074521`	37.0%	0	26.3%	28.3%	8.7%	10.7%	19.5%
K `...23072925`	29.2%	0	17.8%	22.2%	6.9%	11.4%	16.7%

Most of the ignored pixels are both clutter and a disagreement. The two sets are not the same thing: each has a slice the other does not.

The gate, before and after

Median HH, layover minus nominal. Water out. After the ignore, ignored pixels are out too. The full-strip numbers are the ones to compare with V4_CHECK. The tile numbers are what the include rule used: the histogram of the v2 tiles of that strip. Train tiles overlap by half, so a pixel in the overlap is counted twice, and the two medians are not the same quantity.

	near	middle
P, V4_CHECK, every known dry pixel	+1.18	+1.04
P, full strip, ignore applied	**+1.68**	**+1.52**
P, v2 tiles, ignore applied	+1.72	+1.34
K, V4_CHECK	+2.16	+2.62
K, full strip, ignore applied	**+2.46**	**+3.10**
K, v2 tiles, ignore applied	+2.90	+3.32

P clears +1.5 dB on the strip. That is the result the clutter hypothesis predicted, and it is not a change of placement rule. K stays well above the bar. Shadow on these two strips is darker than nominal, and it was already darker in V4_CHECK. The ignore mask did not fix shadow. It took bright non-buildings out of the nominal class.

strip	third	shadow − nominal	interior, eroded 2 m
P	near	−0.92	−1.60
P	middle	−1.18	−1.46
K	near	−1.26	−1.30
K	middle	−1.16	−1.38

Which strips are in the set

A strip is in when, on its v2 tiles, layover minus nominal is at least +1.0 dB in both the near third and the middle third. Water and ignored pixels are out of that contrast. Below +1.5 dB is flagged, not dropped. A tile is dropped only when more than 60% of it is ignored.

Eighteen v2 strips. Fifteen included, three left out. 3042 tiles kept: 2918 train, 95 val, 29 test. 461 tiles on the included strips were dropped for the 60% rule. The three left-out strips account for the other 466 v2 tiles. 3969 − 461 − 466 = 3042.

Left out:

strip	near	middle	why
`20190823134629_20190823134924`	—	—	Six tiles. Layover is only in the far third, and there it is −0.12 dB. The near third has 13 layover pixels and the middle third has none, so there is no median to pass.
`20190822090458_20190822090745`	+1.18	**+0.74**	Middle third under +1.0.
`20190823124221_20190823124528`	**+0.60**	+2.28	Near third under +1.0.

Flagged and kept (under +1.5, at or above +1.0):

strip	near	middle
`20190822074237_20190822074521` (P)	+1.72	+1.34
`20190804111224_20190804111453`	+1.20	+2.54
`20190822133333_20190822133635`	+1.38	+2.18
`20190822085151_20190822085445`	+1.06	+1.62
`20190822134328_20190822134645`	+1.48	+1.44

P is in this list because the include rule reads the tiles, not the full strip. The strip itself is +1.52 in the middle third. The other ten included strips, including K, clear +1.5 on the tiles. Per-strip tables, the ignored fraction of every strip, and the shadow medians are in REPORT_v4.md.

What the kept tiles contain

Classes are `nominal`, `layover`, `shadow`. Shares below are of non-ignored pixels.

split	tiles	nominal	layover	shadow	pixels
train	2918	0.940	0.051	0.0095	5.59×10^{8}
val	95	0.668	0.280	0.052	1.70×10^{7}
test	29	0.568	0.364	0.068	4.77×10^{6}

The split is the v2 west-to-east blocks, unchanged. Val and test are small and much richer in layover than train. A mean IoU on val is not a mean IoU on a random training tile.

Coverage (surfaces per pixel, the severity target) has median 1.0 on nearly every strip. The 90th percentile is 1 or 3. Overlapping layover is a thin tail, not the typical pixel. Ignored pixels keep their coverage value; they are skipped through the mask.

`norm_stats.json` is copied from v2. Same images, same train split.

On a random sample of train tiles the trainer saw 26.5% of pixels ignored, and class frequencies of the rest 0.933 / 0.056 / 0.011. That matches the train row above.

Training

`train.py` reads `classes.json`. On this set it prints 3 classes and the ignore fraction. It was not in the repo before this work, and this commit does not add it.

The overfit at batch 4 does not fit in 4 GB. The kernel killed it (exit 137) before a step was written, and killed it again with `--workers 0`. One overfit of a single batch then ran 200 steps on CPU:

step	loss	nominal	layover	shadow
0	0.915	0.82	0.00	0.00
199	0.028	1.00	0.82	0.00

The loss collapsed and layover rose to 0.82. Shadow printing 0.00 is not a result about that class. The score divides by `max(tp+fp+fn, 1)`, so a class with no pixels in the batch also prints 0.00.

There is no `runs/v4/metrics.json`. The full run (`--out runs/v4 --batch 8`) was not started: batch 4 already does not fit here, and a 50-epoch pass over 2918 tiles does not fit a session on this CPU. There is no `runs/v2_base/metrics.json` in this workspace to put next to it. That comparison needs a GPU.

Not in this commit

The 3042 tiles are not in the repo. v2 is still the only tile release. The scripts that built the masks stayed with the tiles.